

## RED TEAM ATTACK REPORT

# Operation Catch the Hacker

*Attacker reconnaissance and SSH brute force. Real results. Real evidence.***MEL**

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RED TEAM · ATTACK

|        |                                          |              |              |
|--------|------------------------------------------|--------------|--------------|
| Date   | June 30, 2026                            | Attacker IP  | 10.211.55.8  |
| Course | Fullstack Academy Cybersecurity Capstone | Target IP    | 10.211.55.9  |
| Tools  | Nmap, Hydra, Wireshark                   | Capture File | capture.pcap |

## SUMMARY

This report covers the Red Team attack phase of Operation Catch the Hacker, the Fullstack Academy Cybersecurity Capstone project. The objective was to act as an attacker against a target machine on the same local network, find an open service, and break in using a brute force password attack. The full attack was recorded in Wireshark and documented here for handoff to the Blue Team.

The attack succeeded. Hydra recovered valid SSH credentials in under ten seconds. The capture file and this report have been posted to the team Slack channel so Kenny can begin the detection analysis phase.

## LAB ENVIRONMENT

| Machine                         | Role     | OS                       | IP Address  |
|---------------------------------|----------|--------------------------|-------------|
| Kali Linux 2025.2 ARM64         | Attacker | Kali Linux via Parallels | 10.211.55.8 |
| Copy of Kali Linux 2025.2 ARM64 | Target   | Kali Linux via Parallels | 10.211.55.9 |

Both virtual machines ran on the same Apple Silicon MacBook inside Parallels Desktop using the Shared Network adapter. Nmap, Hydra, and Wireshark are included in Kali Linux by default so no additional software was required.

## ATTACK WALKTHROUGH

### Step 1: Both Virtual Machines Running

Both Kali Linux VMs were started in Parallels Desktop. The target VM was confirmed reachable on the local network. Running the `ip a` command on the target confirmed its IP address as 10.211.55.9 on `eth0`.



### Step 3: Wireshark Started to Record the Attack

Wireshark was launched on the attacker VM and set to capture all traffic on the eth0 interface. The capture started before any attack commands were run so that every packet, including the scan and all brute force attempts, would be recorded as evidence for the Blue Team.

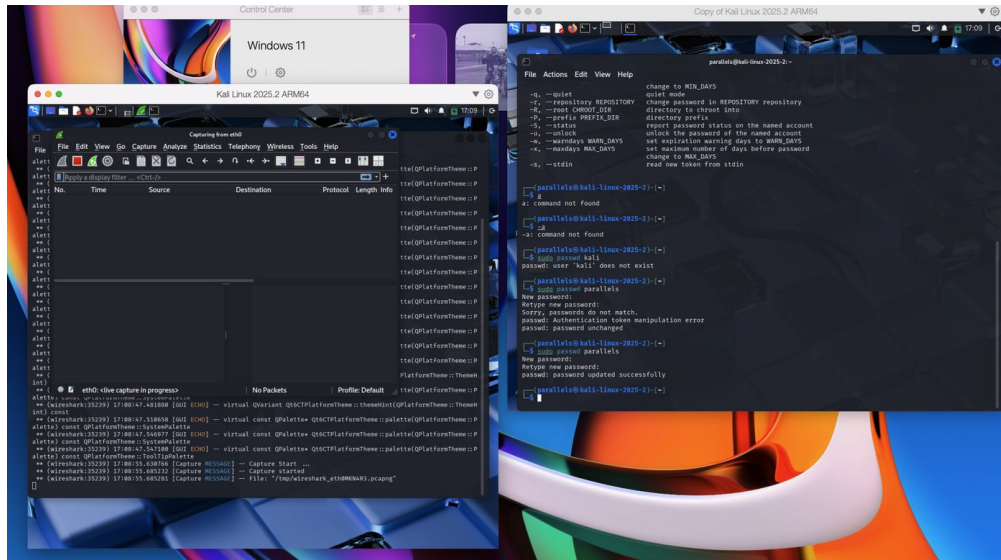


Figure 3: Wireshark actively capturing on eth0. Live traffic visible between attacker and target.

### Step 4: Nmap Reconnaissance Scan

An Nmap service version scan was run from the attacker VM to identify what ports and services were open on the target. The command used was:

```
nmap -sV 10.211.55.9
```

Nmap returned one open port: port 22 running OpenSSH 10.0p2. This confirmed SSH was accessible from the network and gave the service version needed to proceed with a targeted credential attack.

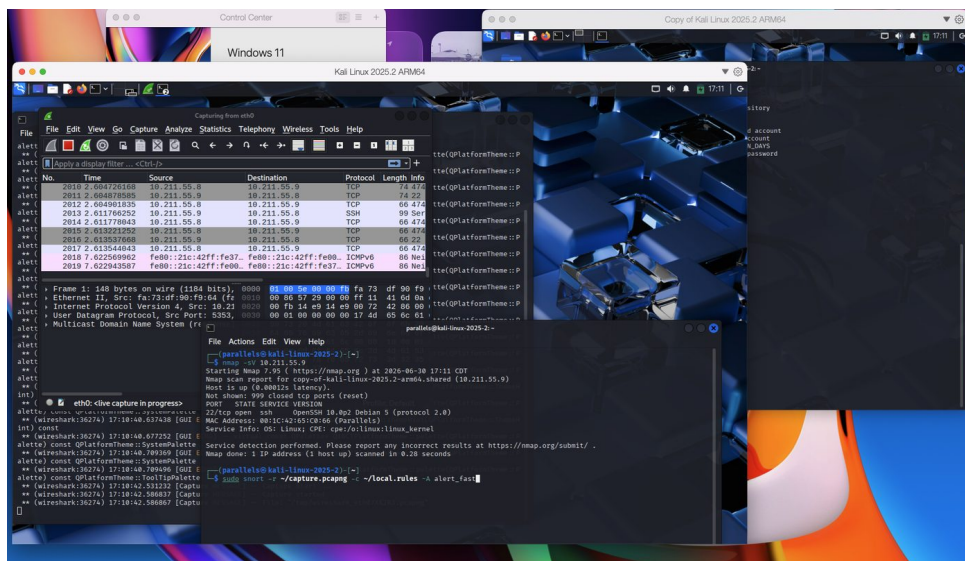


Figure 4: Nmap results showing port 22/tcp open with OpenSSH 10.0p2 on the target.

### Step 5: Password List Prepared

A password wordlist was created using nano containing common weak passwords. This file was saved as passwords.txt and passed to Hydra as the input for the brute force attempt.

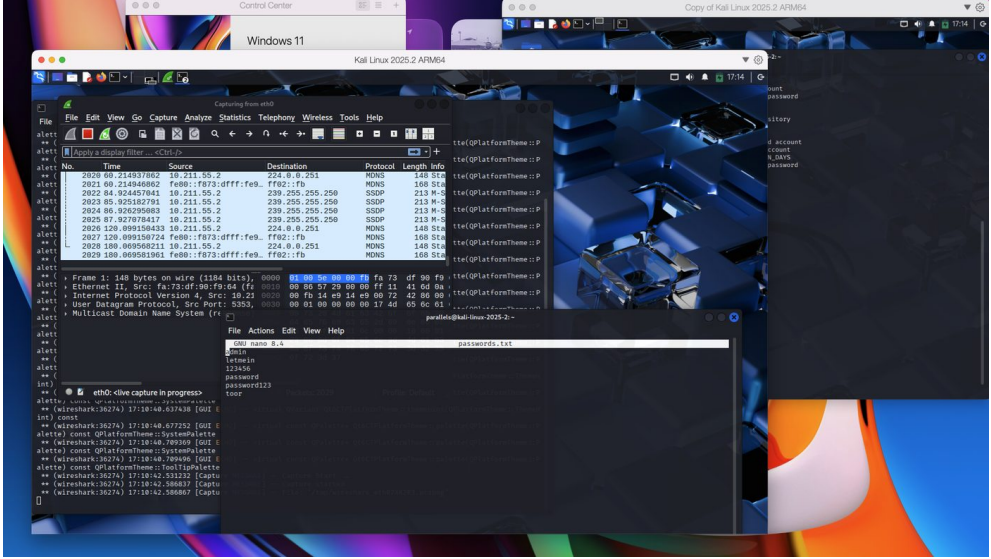


Figure 5: passwords.txt open in nano showing the wordlist used in the attack



## Step 6: Hydra SSH Brute Force Attack

Hydra was used to run a brute force attack against the SSH service on the target. It tried each password in the list against the parallels user account:

```
hydra -l parallels -P passwords.txt ssh://10.211.55.9
```

Hydra found the correct password within seconds. The terminal confirmed a valid login, showing exactly how quickly an exposed SSH service with a weak password can be compromised.

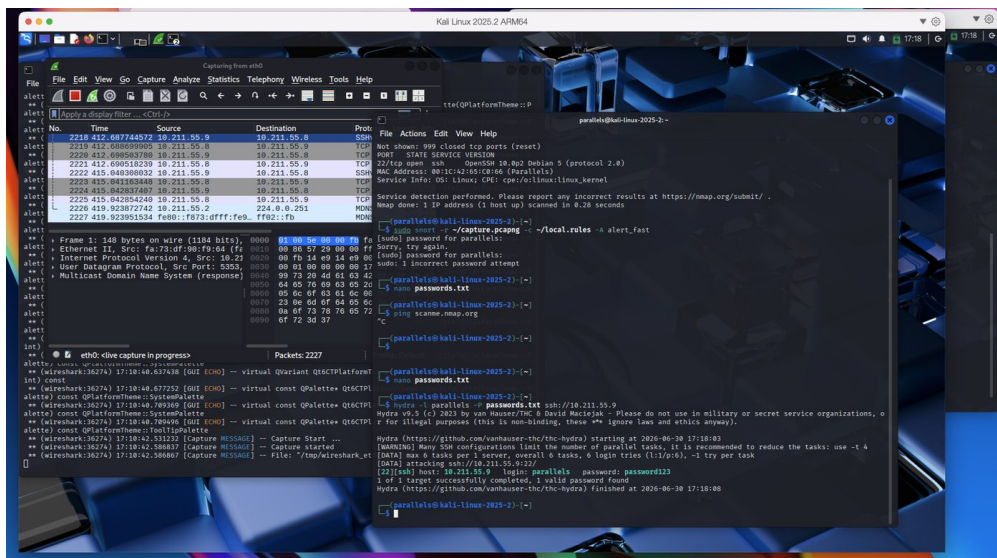


Figure 6: Hydra output confirming the cracked credential. Login: parallels / Password: password123.

**ATTACK RESULT** [22][ssh] host: 10.211.55.9 login: parallels password: password123

## KEY FINDINGS

| No. | Finding                                                                                | Severity |
|-----|----------------------------------------------------------------------------------------|----------|
| 1   | SSH service exposed on port 22 with no access restrictions or IP filtering             | High     |
| 2   | Weak password in use. Hydra recovered credentials in under ten seconds                 | Critical |
| 3   | No account lockout policy detected. Hydra completed all attempts without being blocked | High     |
| 4   | Service banner revealed OpenSSH version 10.0p2, enabling targeted exploitation         | Medium   |

## BLUE TEAM HANDOFF

The following items have been posted to the team Slack channel for Kenny and Mugolie to begin the detection and defense phases.

| Deliverable  | Description                                                      | For   |
|--------------|------------------------------------------------------------------|-------|
| capture.pcap | Full Wireshark packet capture of the attack from start to finish | Kenny |

|                         |                                                           |                 |
|-------------------------|-----------------------------------------------------------|-----------------|
| <b>Screenshots</b>      | Step by step visual evidence of each phase of the attack  | Kenny / Mugolie |
| <b>This PDF</b>         | Written walkthrough, commands used, and findings summary  | Full Team       |
| <b>Screen Recording</b> | Walkthrough video for team presentation while Mel is away | Full Team       |

## CONCLUSION

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The Red Team attack phase was completed successfully using only tools built into Kali Linux. From first scan to confirmed access, the entire attack took under one minute. Nmap identified the open SSH service. Hydra recovered the password through brute force. Wireshark captured every packet as proof of the intrusion.

This exercise demonstrates the real risk of two common misconfigurations: an SSH service open to the network with no restrictions, and a predictable password with no lockout policy. The capture file handed off to Kenny contains the full packet evidence of the attack, including every Nmap probe and every Hydra login attempt, giving the Blue Team everything needed to build detection rules and test firewall hardening.